AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus
E6.1 - Int Trig Substitutions
(A-Level)

Name: ____________________

Class: ____________________

Date: ____________________

Time:	17 min.
Marks:	14 marks
Comments:	

Q1.

(a) Express $7 + 4x - 2x^2$ in the form $a - b(x - c)^2$, where a, b and c are integers.

(2)

(b) By means of a suitable substitution, or otherwise, find the exact value of

$$\int_1^{\frac{5}{2}} \frac{dx}{\sqrt{7 + 4x - 2x^2}}$$

(6)
(Total 8 marks)

Q2.

(a) Show that the substitution $t = \tan\theta$ transforms the integral

$$\int \frac{d\theta}{9\cos^2\theta + \sin^2\theta}$$

into

$$\int \frac{dt}{9 + t^2}$$

(3)

(b) Hence show that

$$\int_0^{\frac{\pi}{3}} \frac{d\theta}{9\cos^2\theta + \sin^2\theta} = \frac{\pi}{18}$$

(3)
(Total 6 marks)